

Machine Learning Approach for Network Intrusion Detection

Task 1 – ML Threat Detection Design Brief

1. Introduction

Modern organizations rely heavily on computer networks to support communication, business operations, and cloud-based services. As these networks continue to grow, they also become more attractive targets for cybercriminals. Attacks such as Distributed Denial-of-Service (DoS), port scanning, brute-force login attempts, and web-based exploits can interrupt services, steal sensitive information, or provide attackers with unauthorized access. Detecting these threats quickly is one of the most important responsibilities of cybersecurity teams. Traditional intrusion detection systems mainly depend on predefined signatures or manually written rules (Sharafaldin et al., 2018). While these methods are effective for recognizing previously identified attacks, they often fail to detect new attack techniques or variations that have not yet been added to their signature database.

Machine learning offers a different approach by identifying suspicious behavior through patterns in network traffic instead of relying only on known signatures. Rather than searching for an exact attack fingerprint, machine learning models learn the characteristics of both normal and malicious network activity from historical data. Once trained, these models can classify new network flows based on the behaviors they have learned, allowing security teams to detect attacks that may not exactly match previously observed examples. This design brief focuses on developing a machine learning solution for network intrusion detection using network flow data similar to the CICIDS2017 dataset. The proposed system is designed to classify network traffic as either benign or malicious while also distinguishing between different attack categories, including DoS, PortScan, and Brute Force attacks. The project combines data preprocessing, feature engineering, supervised machine learning algorithms, and performance evaluation to create a practical threat detection pipeline that can support Security Operations Center (SOC) analysts in identifying malicious activity more efficiently.

2. Problem Definition:

Network intrusion detection is the process of monitoring network traffic to identify activities that may indicate a cyberattack or unauthorized access attempt. In a typical enterprise environment, thousands or even millions of network connections occur every day, making it impossible for security analysts to manually inspect every connection. As organizations continue to expand their digital infrastructure, attackers exploit vulnerabilities through techniques such as Denial-of-Service (DoS) attacks, port scanning, brute-force authentication attempts, malware communication, and other malicious activities. If these attacks are not detected quickly, they can lead to service disruption, data theft, financial losses, and damage to an organization's reputation.

Traditional intrusion detection systems mainly rely on signature-based detection, where incoming network traffic is compared against a database of known attack signatures. Although this method performs well for previously identified threats, it has several limitations. It cannot effectively detect zero-day attacks, new malware variants, or attackers who modify their techniques to avoid matching existing signatures. Maintaining signature databases also requires continuous updates, and any delay in updating those increases the risk of missing newly emerging threats. When designing a machine learning model for cybersecurity, understanding prediction errors is especially important. A false positive occurs when legitimate network traffic is incorrectly classified as malicious. Although this increases unnecessary security alerts and may contribute to analyst fatigue, legitimate users are generally still able to continue their activities after verification. In contrast, a false negative occurs when malicious traffic is mistakenly classified as normal. This type of error is significantly more dangerous because real attacks remain undetected, allowing attackers additional time to compromise systems, steal sensitive information, or move laterally across the network. For this reason, intrusion detection systems usually prioritize high recall, ensuring that as many real attacks as possible are detected while maintaining an acceptable level of false alarms.

Overall, machine learning is well suited for modern intrusion detection because it can identify complex behavioral patterns within large volumes of network traffic, adapt more effectively to evolving attack techniques, and provide security teams with faster and more intelligent threat detection than traditional rule-based approaches.

3. Dataset Analysis

The proposed machine learning model is designed to work with network flow data that follows the structure of the CICIDS2017 dataset, which is widely used in cybersecurity research for intrusion detection. The assignment also provides a synthetic data generator that produces network traffic in a format compatible with CICIDS2017 and UNSW-NB15, making it suitable for developing and testing supervised machine learning models. The generated dataset contains 50,000 network flow records, including 35,000 benign connections and 15,000 attack records, representing an attack ratio of approximately 30%. The attack traffic includes three major categories: DoS, PortScan, and Brute Force, providing sufficient diversity to train and evaluate classification models. The dataset contains both numerical and categorical features. Numerical features include flow duration, forward and backward packet counts, total packets, forward and backward byte counts, total bytes transferred, flow bytes per second, and flow packets per second. These measurements describe the behavior of each network connection and provide valuable information for identifying abnormal traffic patterns.

One of the advantages of this dataset is that it contains labeled examples of both normal and malicious traffic. Each network flow is assigned a label indicating whether it represents benign activity or an attack, along with the specific attack type. This makes the dataset appropriate for supervised machine learning, where models learn the relationship between input features and their corresponding labels.

Although the generated dataset is clean and structured, several data quality considerations should still be addressed before model training. Missing values, duplicate records, incorrect data types,

and inconsistent feature values should be checked during preprocessing to prevent unexpected errors. Outlier analysis is also important because extremely large packet counts or byte values may influence certain machine learning algorithms. In addition, numerical features often have different value ranges, so feature scaling using methods such as StandardScaler or MinMaxScaler can improve model performance, particularly for algorithms that are sensitive to feature magnitude.

Another important consideration is class imbalance, which is common in cybersecurity datasets. Although the generated dataset contains a reasonable distribution of benign and attack traffic, real-world environments usually contain significantly more legitimate traffic than malicious traffic. If a dataset becomes highly imbalanced, machine learning models may become biased toward predicting the majority class. To reduce this risk, techniques such as class weighting or Synthetic Minority Over-sampling Technique (SMOTE) can be applied during training to improve the model's ability to recognize minority attack classes.

4. Model Selection

Three supervised machine learning algorithms are proposed for this project.

- Random Forest will serve as the baseline model because it is reliable, resistant to overfitting, and provides feature importance scores that help explain predictions. It also performs well on structured cybersecurity datasets (Breiman, 2001).
- XGBoost is selected because of its strong predictive performance and ability to handle complex relationships between network traffic features. It generally achieves better detection accuracy than many traditional algorithms and performs well on moderately imbalanced datasets (Chen & Guestrin, 2016).
- Neural Networks are considered as an additional model because they can learn complex patterns in network traffic. However, they require more computational resources and careful parameter tuning compared to tree-based models.

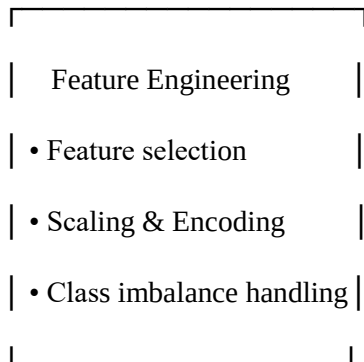
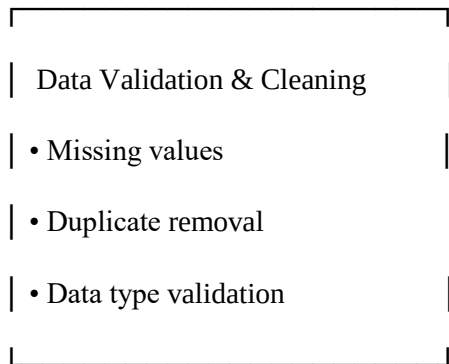
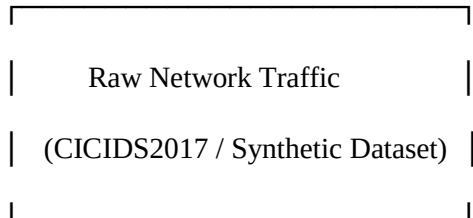
Each model has its own trade-offs. Random Forest offers faster training and better interpretability, XGBoost usually provides the highest detection performance, while Neural Networks have greater learning capacity but require more training time and tuning.

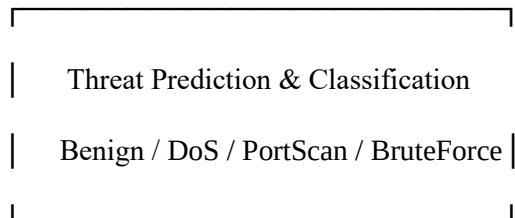
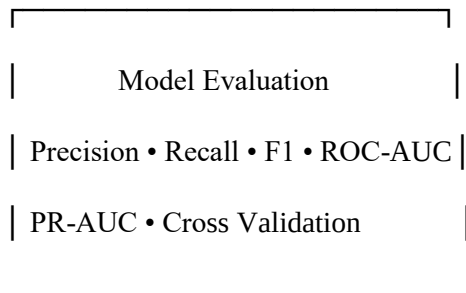
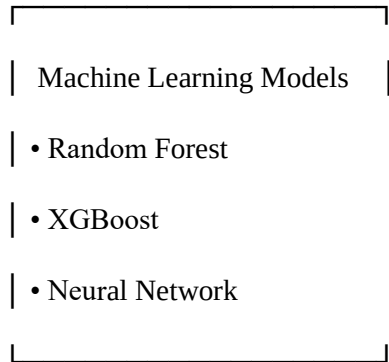
5. Evaluation Strategy:

The models will be evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC. Although accuracy provides an overall performance measure, it may not accurately reflect performance on imbalanced datasets. Precision indicates how many predicted attacks are actually malicious, while recall measures how many real attacks are successfully detected. Since missing an attack can have serious consequences, recall will receive greater importance during evaluation. F1-Score balances precision and recall, while ROC-AUC and PR-AUC evaluate model performance across different classification thresholds.

To ensure reliable results, 5-fold Stratified Cross-Validation will be used during training, followed by evaluation on an independent test dataset that is not used during model development.

6. Machine Learning Pipeline





7. Conclusion

Machine learning provides a practical solution for improving network intrusion detection by identifying malicious behavior based on traffic patterns rather than relying only on predefined signatures. The proposed approach combines data preprocessing, feature engineering, supervised learning algorithms, and comprehensive evaluation to build a reliable threat detection system. By comparing multiple models and prioritizing security-focused metrics such as recall and precision, the system aims to improve detection performance while reducing unnecessary alerts. The resulting pipeline can support security analysts in detecting cyber threats more effectively and adapting to evolving attack techniques.

AI Assistance Note:

I used Claude or ChatGPT for brainstorming, explanation, and language improvement. The final content was reviewed, understood, and edited before submission.